

Imaginary-time Autocorrelation for RTFIM

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1 Random transverse-field Ising model

We consider a periodic chain of length L with Hamiltonian

$$H = - \sum_{i=1}^L J_i \sigma_i^z \sigma_{i+1}^z - h \sum_{i=1}^L \sigma_i^x, \quad J_i \sim U(0, 1), \quad h = h_0/e. \quad (1)$$

The bonds are independent and the transverse field is uniform. The thermodynamic critical point satisfies $\overline{\ln J} = \ln h$. Since $\overline{\ln J} = \int_0^1 \ln J dJ = -1$, it occurs at $h_{0,c} = 1$. For $1 < h_0 < e$, the system is globally paramagnetic but can contain rare, locally ordered segments with very small tunnelling gaps. For $h_0 > e$, the field exceeds every bond and the system is a gapped disordered paramagnet. Thus

$$\begin{cases} \text{Ferromagnetic phase,} & 0 < h_0 < 1, \\ \text{Paramagnetic Griffiths region,} & 1 < h_0 < e, \\ \text{Gapped paramagnetic region,} & h_0 > e. \end{cases} \quad (2)$$

Here $h_0 = e$ is the endpoint of the Griffiths region set by the bounded bond distribution, rather than a second symmetry-breaking transition.

2 Imaginary-time autocorrelation and numerical results

We study the zero-temperature autocorrelation at $j = L/2$,

$$C_j(\tau) = \langle 0 | e^{\tau H} \sigma_j^z e^{-\tau H} \sigma_j^z | 0 \rangle, \quad C_{\text{av}}(\tau) = \overline{C_j(\tau)}. \quad (3)$$

The results use 50 000 disorder realizations for each (L, h_0) . We compare $C = A\tau^{-\alpha}$ and $C = Ae^{-\lambda\tau}$ using equal-weight least squares in $\ln C$. Both models use the same finite positive data at $\tau > 0$; the displayed floor 10^{-8} does not truncate the fits. The labels $1/z$ and $1/\xi_\tau$ in the figures denote α and λ , respectively.

2.1 Field dependence and crossover

Figure 1 shows nearly plateau-like behaviour in the ferromagnetic region. Its very small fitted power-law rates describe slow decay over the available times, rather than paramagnetic Griffiths exponents. Just above criticality, at $h_0 \simeq 1.096$, the curve is approximately straight on logarithmic axes and is well described by a power law with $\alpha \simeq 0.166$. Curvature becomes more pronounced as h_0 increases. Around $h_0 \simeq 1.738$ and 2.089 , neither model reproduces the whole curve well, although the preferred model changes from power law to exponential. At larger fields the exponential is increasingly favoured; preference alone, however, does not imply a close fit.

This intermediate regime reflects the coexistence of several decay scales. Each sample's autocorrelation is a weighted sum of exponentials. A broad distribution of rare-region gaps can produce a power law after disorder averaging, but short-time contributions and finite-size effects bend the curve

Figure 1: Imaginary-time autocorrelations and best fits

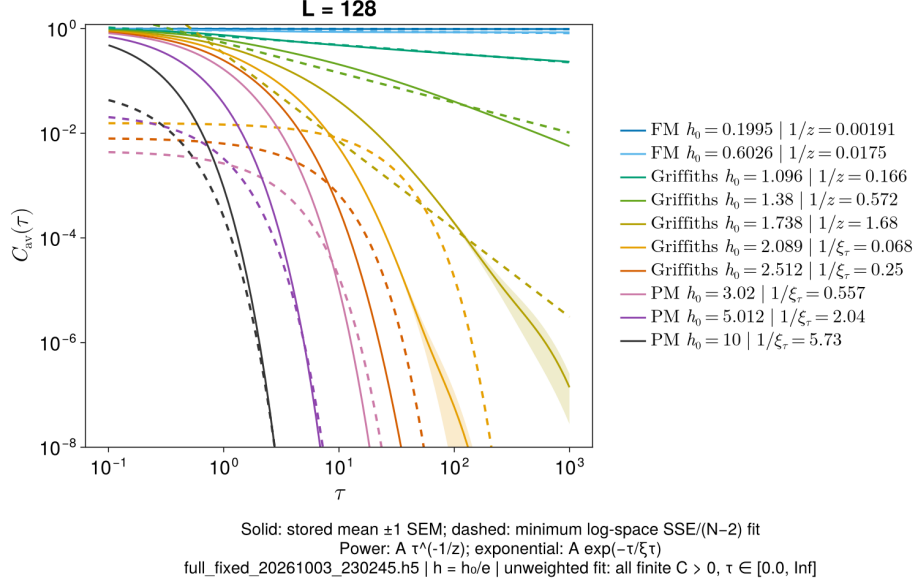


Figure 1: Average imaginary-time autocorrelation at $L = 128$. Colours distinguish fields; solid curves show disorder means, shading indicates one SEM, and dashed curves show the preferred power-law or exponential fit. Legend rates are $1/z = \alpha$ and $1/\xi_\tau = \lambda$. All finite positive data enter the fits; 10^{-8} is only the display floor.

away from that asymptote. As h_0 approaches e from below, the relevant ordered segments become rarer and may be poorly represented by a finite ensemble. Fitting early times and very small late-time tails together can therefore make both simple models inadequate. These are plausible explanations of the observed curvature, rather than causes separately verified by the figures.

The accessible data thus evolve from power-law-like to exponential-like behaviour with increasing h_0 . This does not establish an exponential thermodynamic asymptote inside $1 < h_0 < e$: the true Griffiths tail may require larger systems, more samples or a later-time fitting window.

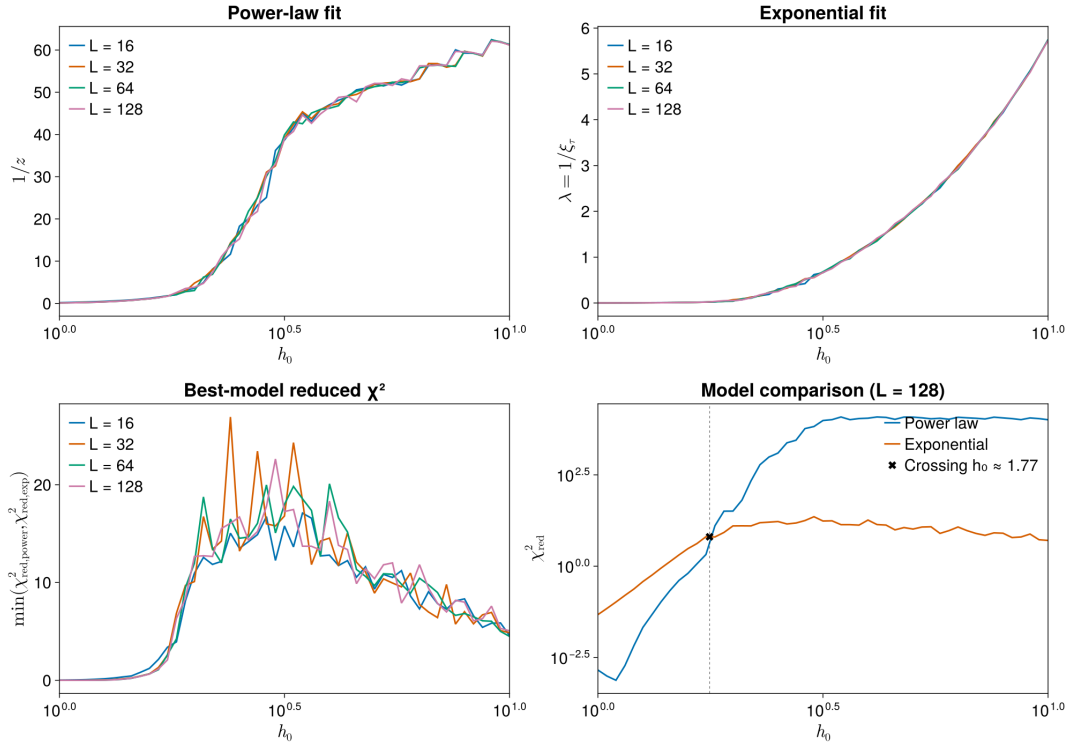
Figure 2 shows increasing fitted decay rates with field. The minimum residual is largest in the intermediate regime and decreases toward large fields, consistent with Fig. 1. The four sizes show similar trends, but this agreement does not establish convergence of the rare-event tail.

2.2 Weighting and interpretation of the fit scores

SEM weighting is not used because the aim is to characterize decay over several time decades. In logarithmic coordinates, propagated SEM weights are approximately $w = (C/\text{SEM})^2$. For $L = 128$, points with $\tau < 1$ carry about 99.40% of the total weight at $h_0 \simeq 1.738$ and 99.56% at $h_0 \simeq 2.089$. Such weighting gives little influence to the long-time behaviour. This dominance depends on field and refers to log-coordinate weights, not $1/\text{SEM}^2$ in the original coordinate. Equal log-space weights balance the decay more evenly, although they do not account for unequal uncertainties or correlations between times.

The plotted “reduced χ^2 ” is actually the unweighted log-space squared residual divided by $N - 2$. Its smaller value identifies the better of two approximations on the chosen window; it is not a standard error-normalized goodness-of-fit statistic. In particular, a value near one has no special statistical meaning.

Figure 2: Imaginary-time fit parameters versus field



full_fixed_20261003_230245.h5 | $h = h_0/e$ | 51 fields, $h_0 \geq 1$
 Unweighted log-space fits; all finite $C > 0$; $\tau \in [0.0, \text{Inf}]$
 Crossings: linear interpolation in $\log(h_0)$, no extrapolation

Figure 2: Field dependence of the power-law rate (upper left), exponential rate (upper right), and smaller fit score (lower left), with colours distinguishing system sizes. The lower-right panel compares both models at $L = 128$. The scores are unweighted log-space residuals divided by $N - 2$; their crossing near $h_0 = 1.77$ marks a change in model preference, not a phase boundary.

At $L = 128$, the scores cross near $h_0 = 1.77$, well below the Griffiths endpoint e . Their relative magnitude therefore cannot determine the phase boundary: an exponential can win within the Griffiths region, and a power law can win for a ferromagnetic near-plateau curve. For this model, $1 < h_0 < e$ and $h_0 > e$ are assigned from the disorder distribution. The fits indicate how clearly the available data display the expected decay. Identifying asymptotic behaviour additionally requires stable long-time slopes and convergence with fitting window, size and sample number.

A JW, Majorana covariance and Pfaffian method

A.1 Quadratic Hamiltonian and ground-state covariance

The Jordan–Wigner transformation defines Hermitian Majoranas

$$a_i = \left(\prod_{k < i} \sigma_k^x \right) \sigma_i^z, \quad b_i = \left(\prod_{k < i} \sigma_k^x \right) \sigma_i^y. \quad (4)$$

They obey $\{\gamma_m, \gamma_n\} = 2\delta_{mn}$ and give $\sigma_i^x = ia_i b_i$ and $\sigma_j^z = (\prod_{k < j} ia_k b_k) a_j$. The Hamiltonian becomes

$$H_q = -ia^T K_q b, \quad (K_q)_{ii} = h, \quad (K_q)_{i+1,i} = -J_i, \quad (K_q)_{1L} = -qJ_L. \quad (5)$$

Here $q = -P$, with spin parity $P = \prod_i \sigma_i^x$; bond contributions are added when matrix entries coincide. For the periodic spin chain, the even ground state uses antiperiodic fermions ($q = -1$). Inserting σ_j^z flips parity, so the intermediate evolution uses periodic fermions ($q = +1$). This sector change is essential.

A singular-value decomposition $K_q = U_q S_q V_q^T$, with singular values $s_{q,\mu}$, determines modes with excitation energies $2s_{q,\mu}$. The initial vacuum energy is $E_0 = -\sum_\mu s_{-, \mu}$. The Gaussian ground state is specified by its Majorana covariance matrix,

$$\Gamma_{mn} = \frac{i}{2} \langle [\gamma_m, \gamma_n] \rangle, \quad \langle \gamma_m \gamma_n \rangle = \delta_{mn} - i\Gamma_{mn}. \quad (6)$$

In the ordering $\gamma = (a_1, \dots, a_L, b_1, \dots, b_L)$, it is

$$\Gamma_0 = \begin{pmatrix} 0 & G_0 \\ -G_0^T & 0 \end{pmatrix}, \quad G_0 = U_- V_-^T. \quad (7)$$

A.2 Spin insertion and imaginary-time evolution

Set $|\psi_j\rangle = \sigma_j^z |0\rangle$. Conjugation by the Jordan–Wigner string changes signs of Majoranas and preserves Gaussianity. The inserted state’s covariance has off-diagonal block

$$G_j = D_a G_0 D_b, \quad (D_a)_{ii} = \begin{cases} -1 & i \leq j \\ +1 & i > j \end{cases}, \quad (D_b)_{ii} = \begin{cases} -1 & i < j \\ +1 & i \geq j \end{cases}. \quad (8)$$

The autocorrelation can therefore be evaluated as the Gaussian overlap

$$C_j(\tau) = e^{E_0 \tau} \langle \psi_j | e^{-\tau H_+} | \psi_j \rangle. \quad (9)$$

In the periodic-fermion evolution basis, define

$$B = U_+^T G_j V_+, \quad D = \text{diag}(e^{-2s_{+, \mu} \tau}), \quad \delta E = \sum_\mu (s_{+, \mu} - s_{-, \mu}). \quad (10)$$

The code’s fast evaluation is the equivalent determinant

$$C_j(\tau) = e^{\delta E \tau} \det R, \quad R = \frac{I + D}{2} + \frac{I - D}{2} B. \quad (11)$$

The energy shift accounts for the different vacuum energies of the two sectors. When R becomes ill conditioned, the code switches to the signed Pfaffian evaluation below to reduce cancellation in small tails.

A.3 Gaussian-overlap Pfaffian

Wick's theorem expresses an even product of Majoranas as the Pfaffian of an antisymmetric contraction matrix. On the ring, the implementation uses its Gaussian-overlap form to retain the change of fermion sector. Choose a Fock reference with occupations $r_\mu \in \{0, 1\}$ that has nonzero overlap with $|\psi_j\rangle$. Relative particle-hole flips allow the odd-parity state to be represented by an even Gaussian chart,

$$S_r = \text{diag}(1 - 2r_\mu), \quad Z = (I - S_r B)(I + S_r B)^{-1}, \quad Z^T = -Z. \quad (12)$$

The code selects the reference by sequentially choosing the more probable occupation, avoiding a singular choice. Write $d_\mu = e^{-2s_{+, \mu}\tau}$ and define diagonal matrices

$$(K_r)_{\mu\mu} = \begin{cases} d_\mu & r_\mu = 1 \\ 1 & r_\mu = 0 \end{cases}, \quad (Q_r)_{\mu\mu} = \begin{cases} 1 & r_\mu = 1 \\ d_\mu & r_\mu = 0 \end{cases}. \quad (13)$$

Then the required $2L \times 2L$ antisymmetric matrix and overlap are

$$W = \begin{pmatrix} Z & -K_r \\ K_r & -Q_r Z Q_r \end{pmatrix}, \quad C_j(\tau) = (-1)^{L(L+1)/2} \frac{\text{pf } W}{\det(I + Z)} e^{\delta E \tau}. \quad (14)$$

The denominator normalizes the Gaussian state, and the fixed sign follows from the displayed block ordering. Computing only $\sqrt{\det \bar{W}}$ would lose the Pfaffian sign. The implementation tracks the sign and $\ln |\text{pf } W|$ separately and combines the normalization and energy shift before exponentiation.

Finally, the mean and SEM are computed over independent disorder realizations. Checks include $C_j(0) = 1$, the decoupled-spin result $C_j(\tau) = e^{-2h\tau}$, and agreement with direct spin diagonalization for small periodic chains.